ENGLISH PROJECT REPORT VERSION

Drone Project - Third Engineering Worksheet

This English version is formatted for graduate application review. It preserves the engineering purpose, design decisions, validation evidence, and individual contribution signals of the original course document while presenting the material in concise English.

Applicant	CHENG-HSUAN LEE
Project	Quadcopter Ball-Transfer Drone
Course	Practice of Mechanical Engineering
Document Type	Worksheet
Primary Role	Team lead and test contributor
Original Source	drone-third-project-worksheet.pdf (6 pages)

Executive Summary

This worksheet supported late-stage test planning, propulsion interpretation, and electronics/debugging follow-up.

It provides process evidence for how the team moved from observed failures toward specific integration decisions.

Applicant Contribution Signals

- Connected propulsion testing with final design choices.
- Recorded troubleshooting priorities as the electronics and control stack became more important.
- Helped turn test results into final strategy.

Worksheet Function

Organize late-stage uncertainties into solvable tasks: propulsion margin, control communication, payload reliability, and final-test procedure.

Technical Decision Table

Constraint	Decision	Why It Mattered	Skill Demonstrated
Late test data	Organized interpretation and action items	Reduced vague debugging	Test analysis
Control stack risk	Tracked signal/debugging tasks	Improved final reliability	Embedded systems
Final strategy	Connected data to execution planning	Prepared evaluation performance	System leadership

How to Read This Evidence

This document is intended to help a reviewer quickly understand the engineering content of the original report in English. The original PDF remains available in the project archive as the source artifact with its original figures, tables, screenshots, and course formatting.